

4. Crypt-Arithmetic Problem

Aim:

To develop a Python program that solves a Crypt-Arithmetic puzzle (e.g., SEND + MORE = MONEY) by assigning unique digits to letters.

Algorithm:

1. Extract all unique letters from the given puzzle strings.
2. Check if the number of unique letters exceeds 10 (invalid puzzle).
3. Generate all possible permutations of digits (0-9) for the unique letters.
4. Check each permutation to ensure leading letters are not assigned zero.
5. Evaluate the mathematical equation.
6. If the equation is valid, print the digit assignments.

```
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import itertools

def solve_cryptarithmic():
    w1, w2, w3 = "SEND", "MORE", "MONEY"
    letters = list(set(w1 + w2 + w3))

    if len(letters) > 10:
        return None

    for perm in itertools.permutations(range(10), len(letters)):
        mapping = dict(zip(letters, perm))

        if mapping[w1[0]] == 0 or mapping[w2[0]] == 0 or mapping[w3[0]] == 0:
            continue

        v1 = int("".join(str(mapping[c]) for c in w1))
        v2 = int("".join(str(mapping[c]) for c in w2))
        v3 = int("".join(str(mapping[c]) for c in w3))

        if v1 + v2 == v3:
            return mapping, v1, v2, v3

    return None

print("Solving SEND + MORE = MONEY:")
res = solve_cryptarithmic()
if res:
    mapping, v1, v2, v3 = res
    print("Solution Found!")
    print(f"{v1} + {v2} = {v3}")
    for k, v in sorted(mapping.items()):
        print(f"{k}: {v}")
else:
    print("No solution exists.")
```

Output:

```
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Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/VISHAL M/Downloads/ai expl.py
Solving SEND + MORE = MONEY:
Solution Found!
9567 + 1085 = 10652
D: 7
E: 5
M: 1
N: 6
O: 0
R: 8
S: 9
Y: 2
>>>
```

Result:

Thus, the Crypt-Arithmetic Problem solved successfully using Python.